CYBER SECURITY

AI SUITE

AI-Powered Threat Detection & Media Forensics Platform

Deepfake Detector · AI Threat Hunter · MITRE ATT&CK Integration

01 DEEPFAKE DETECTOR

AI Vision Forensics · Media Authenticity
OpenCV · PyMuPDF · Groq · Claude · Gemini

Input: Images · Videos · PDFs · Screenshots
Output: AI Forensic Report (.txt download)

02 AI THREAT HUNTER

Isolation Forest · MITRE ATT&CK · IR Playbook
Scikit-learn · Plotly · Anthropic API · Streamlit

Input: CSV · JSON · Syslog · Live Feed
Output: Threat Dashboard · IR Playbook (.txt)

WHY THIS PROJECT MATTERS

Deepfake Crisis	Over **500,000 deepfake videos** circulate online. AI-generated media is used for fraud, disinformation and identity theft — costing billions annually. Traditional detection tools rely on single-vendor APIs with no fallback, failing when one service goes down.
Cyber Threat Surge	Cyberattacks increased by **38% in 2023**. Most organisations lack the tools to detect intrusions in real time. Conventional SIEM systems require months of labelled data before they can detect threats — leaving networks exposed on day one.
Our Solution	This suite directly addresses both challenges with **zero vendor dependency**, **no labelled training data**, and **real-time AI analysis**. MITRE ATT&CK mapping and automated IR playbooks mean any analyst — regardless of experience — can respond effectively within minutes of an alert.
Real-World Impact	Built for immediate deployment: works on any network log format, any media type, with graceful degradation when APIs are unavailable. From raw data to a downloadable forensic report or IR playbook in **under 10 seconds**.

Streamlit	Python 3.10+	OpenCV	Isolation Forest	MITRE ATT&CK;	Plotly
PyMuPDF	Scikit-Learn	Groq API	Claude API	Gemini API	Pandas / NumPy

PROJECT OVERVIEW

The **Cyber Security Suite** is an integrated AI-powered platform combining two tools under a single Streamlit application. Built for the 2026 Cyber Security Competition, it addresses two critical modern security challenges: **detecting AI-manipulated media** and **identifying network intrusions in real time**.

DEEPFAKE DETECTOR	AI THREAT HUNTER
Analyses images, videos and PDFs using multi-provider AI vision APIs to detect GAN-generated faces, splicing, inpainting and document fakery across 7 forensic dimensions.	Ingests network logs and applies Isolation Forest unsupervised ML to detect port scans, brute force, data exfiltration and C2 beaconing — mapped to MITRE ATT&CK.;
Input: Images · Videos · PDFs · Screenshots	**Input:** CSV · JSON · Syslog · Live Feed
Output: AI Forensic Report (.txt)	**Output:** Threat Dashboard + IR Playbook (.txt)

Architecture

Entry Point	Home.py — Streamlit landing with tool selector cards
Core Isolation	sys.modules['core*'] flushed between pages — prevents package collision
AI Key Pool	Groq (Llama Vision) → Claude (claude-3.5-sonnet) → Gemini (gemini-1.5-pro)
Launch	python -m streamlit run Home.py

01 | DEEPFAKE DETECTOR

Analyses **images, videos, PDFs and screenshots** for signs of manipulation using multi-provider AI vision APIs across 7 forensic dimensions. Heuristic fallback runs when no AI key is available.

Analysis Pipeline

Upload	›	Preprocess	›	AI Vision	›	7 Dimensions	›	Report
Image/Video/PDF		**OpenCV · PIL**		**3 providers**		**Forensic analysis**		**Verdict + Score**
Sidebar uploader		Resize, normalise		Key rotation		Structured JSON		Download .txt

7-Dimension Analysis

#	Dimension	What Is Analysed
1	Facial Forensics	Skin texture, eye asymmetry, hair boundary artefacts, geometry
2	GAN Artifacts	Checkerboard patterns, spectral anomalies, upsampling traces
3	Noise & Compression	JPEG DCT blocks, sensor noise distribution, double-compression
4	Lighting & Shadows	Specular highlight consistency, shadow direction, reflections
5	Background & Depth	Semantic coherence, depth-of-field naturalism, boundaries
6	Colour & Chromatic	HSV channel distribution, colour temperature, saturation
7	Document / Text	Font consistency, QR code validity, metadata timestamps

Key Features

AI Key Rotation	Groq → Claude → Gemini — exhausted keys skipped automatically
Fallback	Heuristic analysis (FFT + ELA + noise) runs when all AI keys unavailable
QR Detection	OpenCV QRCodeDetector scans every image; URLs classified for risk
PDF + Report	PyMuPDF scans up to 5 pages; findings exported as downloadable .txt

02 | AI THREAT HUNTER

Applies **Isolation Forest unsupervised ML** to network logs — no labelled training data required. Every detected threat is mapped to **MITRE ATT&CK;** with a full AI forensic report and downloadable incident response playbook.

Detection Pipeline

Ingest CSV/JSON/Live — Upload or stream › **Features** 10 engineered — Log1p + encoding › **Iso Forest** Anomaly score — 0–100 risk › **Classify** 5 threat types — Pattern rules › **AI Report** MITRE + IR — Forensic JSON › **Dashboard** 6 tabs — Export .txt

Feature Engineering — 10 Features

Feature	Encoding	Threat Signal
bytes_sent	log1p(bytes_sent)	Large values signal data exfiltration
bytes_recv	log1p(bytes_recv)	Response volume anomalies
send/recv ratio	log1p(sent/recv+1)	Asymmetric traffic = exfil or scan
dest_port	Raw port number	Unusual ports signal C2 activity
source_port	Raw source port	Ephemeral vs. fixed port patterns
protocol	Label-encoded TCP/UDP	Protocol switches indicate evasion
action	Label-encoded DENY/ALLOW	High deny rate = brute force
conn_rate	Connections per src IP	Frequency anomaly = port scan
unique_ports	Distinct dest ports/IP	Many ports = network scan T1046
duration	log1p(duration) optional	Long sessions = C2 persistence

MITRE ATT&CK; Mapping — 5 Threat Types

Threat Type	Detection Pattern	Tactic ID	Technique
Port Scan	One IP → 50+ ports, tiny bytes, DENY	TA0043 / T1046	Network Service Discovery
Brute Force	High-freq logins to port 22/3389, DENY	TA0006 / T1110.001	Password Guessing
Data Exfiltration	Outbound > 1 MB per event, external IP	TA0010 / T1048.003	Exfil via Unencrypted Protocol
C2 / Suspicious	Beaconing to ports 4444 / 1337 / 9001	TA0011 / T1071.001	Application Layer Protocol
Anomalous Traffic	Statistical outlier, unclassified	TA0007 / T1040	Network Sniffing

Dashboard Tabs

Tab	Content
Overview	Threat timeline scatter, distribution pie, top attacker IPs, heatmap, MITRE bar
Threat Log	Sortable table of all detected threats, risk-coloured, CSV export
AI Analysis	Select any threat → full AI forensic report + MITRE + risk gauge
Incident Response	AI-generated IR playbook → download as .txt for SOC handoff
Network Graph	Force-directed graph: red nodes = attackers, red edges = attack paths
Live Feed	@st.fragment 5s refresh from live_feed.csv — full page never reloads

AI FORENSIC ANALYSIS & INCIDENT RESPONSE

When a threat is selected, the AI generates a 12-section forensic report using Claude AI (with key rotation and rule-based fallback). The IR playbook is ready to hand to any SOC analyst.

AI Threat Report — 12 Sections

Section	Content
Severity & Confidence	CRITICAL / HIGH / MEDIUM / LOW + AI confidence 0–100%
Threat Title	Human-readable name, e.g. 'Coordinated Port Scan — 185.220.x.x'
MITRE ATT&CK;	Tactic ID + Name → Technique ID + Name → Sub-technique
Kill Chain Phase	Reconnaissance / Weaponisation / Delivery / Exploitation / C2 / Exfiltration
Executive Summary	2–3 sentence plain-language description of the incident
Evidence Analysis	Specific packet-level evidence supporting the classification
Impact Assessment	Business risk: data loss, system compromise, service disruption
Threat Actor Profile	Type · Sophistication (Low / Medium / High / Nation-State) · Motivation
Indicators of Compromise	IPs, ports, byte patterns, timing signatures as IOCs
False Positive Assessment	Likelihood this is benign traffic with explanation
Related Threat Types	Other attacks commonly paired with this technique
Analyst Notes	Caveats, confidence limitations, edge cases

Incident Response Playbook

Phase	Timeframe	Actions
Immediate Actions	0 – 15 min	Block IP at firewall, isolate endpoint, preserve logs
Investigation Steps	15 – 60 min	SIEM log pull, EDR check, lateral movement hunt
Containment & Eradication	1 – 4 hrs	Revoke credentials, patch exploited service, clean malware
Long-term Remediation	Days	Deploy detection rule, update policy, lessons learned

Live Feed — Real-Time Demonstration

Generator	live_log_generator.py writes to live_feed.csv at 0.5–1.5s intervals
Attack Injection	Port Scan / Brute Force / Data Exfil / C2 beacon injected every 20–45s
Refresh Method	@st.fragment(run_every='5s') — only the Live Feed panel re-runs
Start	Terminal 1: python -m streamlit run Home.py | Terminal 2: python live_log_generator.py

TECHNOLOGY STACK & COMPETITION DEMO SCRIPT

Full Technology Stack

Library	Version	Purpose
Streamlit	1.39+	Web UI — multi-page, fragments, session state
Scikit-learn	1.4+	Isolation Forest unsupervised anomaly detection
OpenCV	4.x	Image preprocessing, QR detection, video frame extraction
Plotly	5.x	Interactive scatter, bar, pie, heatmap, network graph, gauge
Anthropic / Groq / GenAI	Latest	Claude · Llama Vision · Gemini — AI vision APIs
PyMuPDF (fitz)	1.24+	PDF page rasterisation at 2x zoom
Pandas / NumPy	2.x	Data ingestion, feature engineering, statistics
Pillow (PIL)	10.x	Image loading, format conversion, preprocessing

2-Minute Demo — Deepfake Detector

Step	Action
1	Home → Launch Deepfake Detector → Sidebar: Analyze Demo
2	Point to verdict banner: FAKE / confidence % — AI manipulation type shown
3	Expand AI Forensic Report → primary indicators + 7-dimension breakdown
4	Upload a real photo → shows AUTHENTIC with different evidence signals
5	Upload a PDF → multi-page scan with QR code detection result
6	Click Download Report → show the .txt forensic output

2-Minute Demo — AI Threat Hunter

Step	Action
1	Home → Launch AI Threat Hunter → Use Sample Data
2	Overview tab: timeline — red dots are attacks, dot size = data volume
3	Heatmap: attacks cluster at 2–4 AM — classic attacker behaviour
4	AI Analysis tab: select highest risk alert (Data Exfiltration, 90+/100)
5	MITRE mapping: TA0010 → T1048.003 · Kill Chain: Exfiltration phase
6	Incident Response tab → download IR playbook as .txt
7	Sidebar → Live Feed mode → Terminal 2: python live_log_generator.py
8	Live Feed tab: new attack appears every 20–45s, panel refreshes every 5s